

Lesson 7 — Fast, Slow, Forward, Back

Matches LEGO Teacher Portal lesson: "Elephant's Play Day".

Objective (I can...): I can make the motor move in different ways — faster, slower, forward, and back.

CS concept: Investigating motor movements (speed, direction, and how long) · **Builds on:** Lesson 6 (start/stop a motor).

AI Connection (2 min, read aloud): A delivery robot speeds up on a clear path and slows down near people — it controls its motors carefully. *Ask the group:* "When should a robot go slow instead of fast?"

Materials: LEGO CS & AI kit + device per pair, a short "road" strip of tape on the table or floor. **Model reuse:** Hardware machine from Lesson 6 — code only.

60-minute flow - Warm-up (10): "Fast and slow" — kids march in place super fast, then slow-motion, then backward on "reverse!" Feel speed and direction with their bodies. - **Concept + demo (8):** Show the motor-speed picture and the direction picture. Run the motor fast, then slow, then the other way, so kids see the change. - **Build/Code (25):** Pairs explore motor movements: code the motor to go fast, then slow, then reverse, watching each time. On the "road" strip, they try to make the machine move a short way and stop near a line. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Make the motor do "fast, then slow, then back" in one program; guess what the machine will do before running. - **Share + tidy-up (10):** Pairs show their favorite motor move and name it (fast / slow / back). Model stays built.

Pair roles: Driver codes the motor moves, Navigator calls out "faster, slower, back" and watches the result. Swap at the wiggle break. (*Trio rotates code/call/watch.*)

Differentiation: *Support* — change only speed (fast vs. slow). *Extension / fast-finisher* — make the machine travel to a line and stop. *Reteach* — change one setting at a time and run after each change so the difference is clear.

Exit check: Pair can make the motor change speed and direction on purpose. **At-home (optional):** Watch a fan or toy that speeds up and slows down; tell a grown-up what makes it change. ~5 min.
